

**No paper**

Google has dropped the long standing idea of buying a newspaper said CEO Eric Schmidt

**PowerMeter**

Google recently announced partners for its PowerMeter utility, designed to save energy by generating energy usage graphs

Industry Connect

Digit in a freewheeling interview with Prasad Ram, head of the Google R&D centre in Bangalore, India

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What kind of innovative technologies is Google working on?

The emphasis is on building universal products. In a market such as India, universality is the key. However, we have to keep the local flavour in mind. The product is designed for a very large audience, but fulfils the needs of the local audience. India has a very unique position. We are a billion people in a democratic market (unlike China), and the diversity in this market is unparalleled. Nowhere else will you find as many differences in education, levels of income, languages, cultural pursuits, you name it. In urban settings, people are interested in social networking, entertainment, YouTube and so on. You go to tier II towns and you'll see a huge emphasis on education and classifieds because these impact their lives directly. When you go a step further to tier IV towns and villages, language becomes a problem. The attitude is that computers and internet is for English speaking people and we don't need them.

For example, in search we suddenly saw this huge spike of searches for "RBE" coming from a particular region. It turns out this stands for Railway Board Exam. Now our job is to provide a great search experience. Today if you type GOI, Google knows in India it means Government of India. Similarly, we intend to provide a great user experience to anyone looking for a particular piece of information. So if you typed RBE, you get the correct result. Also, it's common to search for Indian words on Google. But the way we spell Indian words differs. *Bachchan* may be spelt in so many ways,



As head of the Google R&D centre in Bangalore, India Prasad Ram is responsible for working with the India engineering teams on the development of innovative Google products and technologies

for example. The expectation of the user is that Google should understand all this; give me what I mean, not what I ask. So we want to give a great user experience and localise search along different dimensions. Localised by geography, economic status, language, and so on.

What is Google's outlook for the Indian market?

The market itself is extraordinarily exciting. It's relatively untapped in terms of internet usage. We have extraordinary talent in the country. For some good reason, all of us have a huge IT bias. As a result, we have the people who want to do good work.

What about the mobile platform?

Undoubtedly, the mobile platform is important. The interesting part of the mobile platform is currently more in the SMS space. This is because though we may have more than a hundred million phones in India that are data enabled, sadly, people don't browse significantly on their mobile

devices. So as for our SMS search we're doing a lot of work on its back end. There are a lot of challenges to this. Primarily, because the results are sent to the user for a particular query, it's almost like hitting "I'm feeling lucky" each time. You get one result and it should be the right result for you. Also, in SMS search, I can't know whether the user liked my result; more importantly whether he got anywhere as a result of it.

Any innovations?

In mobile search, one of the biggest challenges is understanding accents. Even our grammar constructs are influenced by the underlying native language we're more familiar with. Building a good recognition model is very important here. You might call in and ask for something which may be a native word combined with other English words. You may think you're speaking English, but there are few native words creeping in. But phone search is just about now gaining popularity in the country and these are the technical challenges we're facing.

Any research on natural language processing?

At Google, we collaborate on almost everything other Google centres across the world. For NLP, some of our best minds are based in New York and Mountain View. Of course, some of the linguistic problems we face here are uniquely Indian, but once we solve the problems, we solve them generally enough that they should apply anywhere; even the US where there is a fairly cosmopolitan mixture of people as well.

Looking forward?

If you look at the affordability of PCs lately, and the penetration, it's highly skewed. Some data said 12 per cent of Indian households can afford PCs, but only two per cent have them. There are 200,000 cyber cafes in India so accessibility is there. If I live in Nashik and I'm looking at going for a movie, chances are listings may not be available online. We're looking at changing that. That way the relevance of the internet goes up. The Map Maker product is a step towards this and it's got spectacular response. It was an India problem that inspired us to come up with this idea, but now it's a global product, launched worldwide. On the mobile platform, we have launched SMS channels where we've got a very positive response on it. People are creating and subscribing to various channels. They're actively publishing lots of information through these channels, so it's quite exciting. As the awareness of these channels goes up, every small community can stay informed and in sync by creating community specific channels. Communities here could be described as your apartment complex or it could even be the chess community of Mumbai. 